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Lab 3: Robot Manipulation – De-palletizing Using a Universal Robots Collaborative Manipulator

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1 Task Description

The objective of this laboratory session is to program and execute a complete de-palletizing manipulation task using a Universal Robots collaborative manipulator within the PolyScope programming environment.

The task consists of autonomously handling identical workpieces arranged in a palletized configuration. The robot must sequentially approach each pallet cell, detect and grasp the workpiece using a vacuum gripper, lift it safely from the pallet, and place it at a designated drop-off location. This process emulates an industrial de-palletizing scenario commonly encountered in logistics and manufacturing lines.

The lab is structured into three incremental levels, each increasing the complexity and autonomy of the robotic system:

- **Level 1:** Basic de-palletizing sequence. The robot follows a fixed and predefined order to pick parts from known pallet positions and place them at a fixed unloading location.
- **Level 2:** Layer-wise de-palletizing. The robot iterates through multiple pallet layers, adjusting the vertical approach height dynamically as layers are removed, while placing parts into predefined target positions.
- **Level 3:** De-palletizing with sensing and disturbances. A tool-mounted contact or vacuum sensor is introduced. Some pallet cells may be empty, requiring the robot to verify part presence before grasping and to skip missing positions automatically.

By the end of this laboratory task, the robot executes a fully autonomous de-palletizing cycle: iterating through pallet coordinates, detecting part presence, safely extracting objects, and placing them at the target location while reacting robustly to missing parts and environmental disturbances.

2 Program Breakdown

The developed PolyScope program controls the Universal Robots manipulator to execute an automated de-palletizing process consisting of pallet scanning, object grasping, lifting, and placement. The program makes extensive use of Cartesian linear motions (`MoveL`) combined with digital I/O signals to actuate the vacuum gripper, monitor contact feedback, and ensure safe manipulation. The logic is structured to comply with the laboratory requirements for Levels 1, 2, and 3, enabling repeatable and collision-free de-palletizing operations.

Note: A single tool frame is used throughout execution, corresponding to the vacuum gripper mounted at the robot flange. Approach poses (`up*`) and grasp poses (`pick*`) were defined relative to the pallet coordinate system, while placement poses (`place*`) correspond to the unloading station reference frame.

Motion Sequence

The motion sequence is structured around safe approach, grasp, lift, and placement trajectories:

For clarity, symbolic pose names are used to describe approach and operation points. These names correspond to actual taught waypoints in the PolyScope program.

- **Pallet Points:** (`uppallet`, `pickpallet`) define the approach and grasp positions for each pallet cell.
- **Lift Points:** (`uppallet`) ensure vertical clearance before horizontal motion.
- **Placement Points:** (`upplace`, `place`) define the unloading location for the extracted objects.

All movements are executed using `MoveL` commands to ensure predictable Cartesian paths. Each motion completes fully before the next action is initiated to maintain operational safety.

I/O Operations

- **Outputs:**
  - `vacuumOn` – Digital output activating the vacuum gripper.
  - `vacuumOff` – Digital output releasing the vacuum.
- **Inputs:**
  - `vacuumOK` – Confirms successful object attachment.
  - `toolContact` – Detects contact or pressure at the tool (used in Level 3).

Waiting and Synchronization

Synchronization between robot motion and gripper actuation is critical for safe de-palletizing. The program ensures reliable execution using:

- Motion completion checks inherent to `MoveL` execution.
- Conditional waits on sensor feedback (e.g., vacuum confirmation).
- Short delays following gripper activation to stabilize grasping.

These synchronization steps prevent premature lifting, ensure stable gripping, and reduce the risk of dropped objects.

Program Flow (Pseudocode)

Task Parameters

The parameters used in this de-palletizing task include robot speed configuration, pallet grid coordinates, tool frame definition, and digital I/O mappings for the vacuum gripper and sensors.

Joint and Cartesian Positions

The robot uses predefined Cartesian poses for:

- **Pallet Handling:**
  - `uppallet` – safe vertical approach above pallet cells.
  - `pickpallet` – grasp pose at pallet cell.
- **Placement Handling:**
  - `upplace` – safe approach above unloading location.
  - `place` – final placement pose.

All poses were recorded using the teach pendant and adjusted during execution testing.

Tool Frame

A single tool frame corresponding to the vacuum gripper was defined. This frame ensures correct alignment between the robot flange and the object surface during grasping and placement.

Robot Speed

All motions use conservative velocity and acceleration limits to comply with collaborative robot safety requirements. Initial tests were conducted at reduced speed before increasing to nominal values.

Algorithm 1 De-palletizing Procedure using a Universal Robots Manipulator

Initialize robot and load tool configuration
Set nominal robot speed and acceleration limits
Initialize digital inputs and outputs
Move robot to a safe home pose
Display "System Ready – Starting De-palletizing Cycle"
for each pallet layer **do**
  for each pallet cell in layer **do**

    // — **APPROACH PALLET CELL** —
    Move above pallet cell: `MoveL(uppallet)`
    Move to grasp pose: `MoveL(pickpallet)`

    // — **GRASP OBJECT** —
    Activate vacuum gripper
    Wait until `vacuumOK == true`

    // — **LIFT OBJECT** —
    Retract vertically to safe height: `MoveL(uppallet)`

    // — **PLACE OBJECT** —
    Move above placement location: `MoveL(upplace)`
    Move to placement pose: `MoveL(place)`
    Deactivate vacuum gripper

    // — **RETRACT** —
    Move back to safe height: `MoveL(upplace)`
  end for
end for
Display "De-palletizing Completed Successfully"
Return robot to home pose

Digital I/O

The digital inputs and outputs used are:

- **Outputs:**
  - Vacuum control signals (on/off)
- **Inputs:**
  - Vacuum confirmation sensor
  - Tool contact or pressure sensor

All signals were configured within the PolyScope I/O interface.

Safety Considerations

The Universal Robots manipulator is a collaborative system; however, strict safety measures were enforced throughout the experiment. All trajectories were validated in simulation before execution. Vertical approach and retreat motions were enforced to minimize collision risks with the pallet and surrounding structures.

Robot speed and acceleration were limited during initial testing, and emergency stop functionality was verified prior to execution. All tool actions were synchronized with sensor feedback to prevent unsafe lifting or accidental object release.

Clear approach poses and conservative motion planning ensured safe and repeatable operation across all de-palletizing cycles.

Execution Results

The robot successfully executed the complete de-palletizing routine, removing objects layer by layer from the pallet and placing them at the target location. All objects were grasped securely using the vacuum gripper and transported without loss or collision.

The system demonstrated stable performance under repeated execution. When sensing was enabled, empty pallet positions were correctly skipped without interrupting the workflow.

Overall, the task confirmed the reliability of the programmed logic and the suitability of collaborative robots for structured de-palletizing operations.

Improvements

Several enhancements could further optimize the system:

Reducing cycle time:
Blended motions and higher speed limits (within safety constraints) could reduce total execution

time.

Adaptive pallet handling:
Automatic detection of pallet dimensions and object height would increase flexibility.

Advanced sensing:
Integrating vision-based detection could further improve robustness in unstructured environments.

Error recovery:
Adding retry logic for failed grasps would improve industrial reliability.



Figure 1: Before



Figure 2: After